

Ping-Jhao Hsu

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Education

National Cheng Kung University (NCKU)

Bachelor of Science in Engineering Science

Sept. 2022 – Present

Tainan, Taiwan

Relevant Coursework

Mathematics: Discrete Mathematics, Linear Algebra, Engineering Mathematics

Computer Science: Programming, Data Structures, Algorithms, Logic Design, Theory of Computation, Database Systems, Network Security, Numerical Methods, Computer Organization & Assembly, Microprocessors & Windows Programming, Image Processing & Computer Vision, Deep Learning, Optimization and Reinforcement Learning

Electrical Engineering: Electric Circuits, Electronics, Electronics Laboratory, Robotics Principles, Control Systems

Mechanical Engineering: Engineering Graphics & Simulation, Engineering Mechanics, Engineering Dynamics, Thermodynamics, Fluid Mechanics

Other: Introduction to Carpentry

Projects

LightPOV | C, C#, JavaScript, SolidWorks, KiCad

July 2024 – Aug. 2024, Oct. 2025 - Present

- Engineered a wireless LED control system on ESP32, utilizing dual-core processing to synchronize real-time C-based firmware with Wi-Fi communication via a Node.js web interface.
- Fabricated 3D-printed internal structures to integrate and secure hardware components.
- Created a C# GUI editor to arrange HSV lighting effects and music nodes, exporting scripts via JSON.

Tatsujin | Verilog, PYNQ-Z2

June 2024

- Implemented a Tatsujin game clone on a PYNQ-Z2 FPGA using an LED matrix and Verilog.
- Designed Finite State Machines (FSM) for note detection, scoring logic, and game state management.
- Fabricated a custom controller and developed animation rendering logic.

DDQN Space Survival Shooter (Reinforcement Learning) | Python, PyTorch

June 2025

- Implemented a Double DQN Game Agent using PyTorch, incorporating Replay Buffer, ϵ -greedy decay, and Target Network synchronization.
- Designed state representation by abstracting the game screen into a 51-dimensional vector and engineered the reward function.
- Integrated TensorBoard to visualize loss, reward, and ϵ curves for performance analysis and debugging.

Virtual-Real Sync Lighting System | Unity3D, Meta Quest 3

Mar. 2025 – Present

- Developed immersive lighting environments in a Mixed Reality (MR) environment using Meta Quest SDK and Unity.
- Implemented intuitive light parameter adjustments using hand-tracking gestures.
- Programmed C scripts in Unity to transmit real-time commands via WebSocket, synchronizing physical lighting hardware.

Skills & Certificates

Embedded System: Arduino, ESP32, PCB design

Development: Python, C/C++, Java, C#, SQL

Software: SolidWorks 2018, Fusion360, EasyEDA, Kicad

Prototyping: woodworking, 3D printing(FDM/SLA), soldering, laser cutting

Leadership & Activities

Maker Space

Sept. 2022 – Present

Member

- Leveraged proficiency in 3D printing, circuit soldering, and Arduino to develop rapid prototypes.
- Initiated and completed a summer research project "LightPOV" (June 2024 – Aug. 2024).

Freshman Orientation Camp (ES x PT x OT Depts.)

Sept. 2023 – Sept. 2024

General Coordinator / Team Leader

- Directed a team of 50+ members; orchestrated pre-camp meetings and training sessions.
- Bridging three academic departments, external vendors, and university administration.

25th / 26th / 27th Engineering Science Camp

Jan. 2023 – Feb. 2025

Activity Director / Staff / Team Leader

- Allocated responsibilities for 50+ staff members and directed camp logistics to ensure high participant engagement.
- Spearheaded organizational reforms to modernize legacy procedures, significantly boosting team efficiency and morale.

Engineering Science Student Association

Sept. 2023 – Sept. 2024

Head of Activities Department

- Orchestrated major departmental events (e.g., Christmas Karaoke, Farewell Party), fostering a cohesive student community.